

Ranking Transfers, Thresholds Don’t: Prior-Corrected Test-Time Adaptation for Audio Deepfake Detection

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Abstract—Audio deepfake detectors generalise poorly across corpora despite near-perfect in-domain accuracy. We show the failure is one of *operating point*: a fine-tuned self-supervised detector’s *ranking* (ROC-AUC) largely transfers cross-corpus while its decision *threshold* does not, so unlabeled target audio can repair the operating point at test time. We further show that confident-quantile test-time adaptation (TTA) silently assumes a *class-balanced* target pool — an assumption that holds where TTA is measured and essentially nowhere else, in either direction: the standard ASVspoof2021-DF benchmark is 97.2% spoof, In-the-Wild is 37.2%. On the official DF eval (400,435 trials) the published symmetric-quantile configuration destroys the operating point of three independently trained state-of-the-art released checkpoints, driving accuracy from 98.8–99.1% to 62–65%; three of four modern TTA baselines (Tent, SHOT, ETA) degrade them too, each through its own version of the same assumption, while SAR avoids damage only by reverting its own updates. Correcting the pseudo-label budget with a label-free estimate of the target prior ($\hat{\pi}=0.9715$ vs. a true 0.9708) instead preserves accuracy and improves EER on all three, beating the symmetric budget by 0.79–1.82 points and the source model by up to 27.9% relative. Because EER and AUC both move, no re-thresholding rule accounts for it: the one-line median-threshold control that looks competitive on balanced pools loses 40 accuracy points at this benchmark’s real prior. Two label-free monitors — rank agreement with the frozen source model, and prediction-rate drift — separate the adaptations that damaged a checkpoint from those that did not.

Index Terms—audio deepfake detection, anti-spoofing, test-time adaptation, label shift, cross-corpus generalisation, EER.

I. INTRODUCTION

Self-supervised (SSL) front-ends such as wav2vec 2.0 and XLS-R, paired with lightweight back-ends, reach very low in-domain error on standard anti-spoofing benchmarks [1]–[3]. Deployed on audio from a different source—new microphones, codecs, languages, or generators—accuracy drops steeply, a well-documented generalisation gap [5], [6], [17].

We start from a diagnostic measurement: a fine-tuned XLS-R detector evaluated on an unseen corpus keeps a high **ROC-AUC while its accuracy at a fixed threshold collapses**. It still *ranks* clips correctly; its operating point is mis-calibrated. Because ranking transfers, the most-confident predictions on unlabeled target audio are trustworthy enough to adapt on at test time.

That observation is the starting point of a family of methods, not a method. The step every member of the family takes—label the extreme quantiles of the target scores and self-train on them—carries an assumption that is invisible on the benchmarks the field uses and fatal on the pools it deploys to: that the target pool is roughly class-balanced. Take the top and bottom 30% of a 97%-spoof pool and the “confidently bona fide” bucket is mechanically made of low-scoring *spoof*. Self-training then teaches the detector that a great deal of spoof is genuine. This is not hypothetical: it is what happens to a released state-of-the-art checkpoint on the official ASVspoof2021-DF protocol (Table I).

Contributions. (i) The ranking–calibration asymmetry as the mechanism of cross-corpus failure, over 369 third-party cells and tested directly: a proxy $\mathcal{A}$ -distance *rises* after adaptation, so the effect is operating-point repair, not learned invariance. (ii) The class-prior assumption hidden in confident-quantile TTA and in three of four modern TTA baselines, measured on the field’s standard benchmark at its real prevalence, where it destroys the operating point of three released SOTA checkpoints. (iii) A label-free correction—shift estimation of the target prior, used to split the pseudo-label budget—which removes the damage and beats the symmetric budget on every checkpoint, threshold-free. (iv) Two label-free monitors that separate damaging adaptations from safe ones, including our own published configuration.

Related work. SSL front-ends with AASIST or attentive back-ends dominate recent benchmarks [1]–[4]. Train-time domain generalisation aligns source domains [5] or runs gradient reversal over source and target jointly [13]; both act before deployment. In TTA, entropy minimisation (Tent) [7] is the reference recipe, and its instability prompted reliable-sample filtering (ETA/EATA) [8], sharpness-aware entropy with a recovery scheme (SAR) [9], and information maximisation with self-labelling (SHOT) [10] — we compare against all four. Batch-statistics recalibration [14] is a no-op for LayerNorm. The prior estimator is black-box shift estimation [11]; the proxy $\mathcal{A}$ -distance is from [15].

II. METHOD

Let x be a waveform with label $y \in \{0, 1\}$ (fake= 1) and let f_θ produce $s_\theta(x) = P_\theta(y=1 \mid x)$. We are given a source

model, its labelled training corpus $\mathcal{S}$, and an *unlabelled* target pool $\mathcal{T}$. No target label is read at any point.

What breaks under corpus shift. Ranking quality $\text{AUC} = \Pr[s_\theta(x^+) > s_\theta(x^-)]$ is threshold-free; accuracy at a fixed τ is not. Empirically AUC degrades far less than accuracy at $\tau=0.5$ under source→target shift: across 369 third-party evaluation cells the accuracy a model’s ranking supports but its shipped threshold does not deliver averages 7.29 points. The deployable object is therefore the score *order*, and the broken object is the operating point. Everything below follows from taking that literally.

(a) The target prior, label-free. An operating point is a statement about a prior, so we estimate the prior rather than assume it. Black-box shift estimation [11] gives $\hat{p} = M^{-\top}q$, where $M[y, \hat{y}] = P_{\mathcal{S}}(\hat{y} | y)$ is the source confusion matrix measured on the model’s own labelled training data and q is the observed target prediction histogram. M and q read source error structure and target *prediction rate* only—never the shape of the target score distribution, which is precisely what mis-calibration corrupts, so BBSE still works where a score-histogram estimator has a degenerate histogram to work from. On a well-separated checkpoint the choice of estimator does not matter: at the real DF prior a two-component GMM, an EM estimator that needs only the source class balance [12], and BBSE agree to within 0.02 and produce effectively the same tail split (mean absolute error over a 0.5–0.97 sweep: 0.028, 0.040, 0.041). What matters is estimating the prior at all rather than assuming it is $\frac{1}{2}$.

(b) Prior-corrected confident tails. The standard rule takes the bottom q and top q of the target scores as pseudo-real and pseudo-fake. We split the same total budget $2q$ by the estimated prior,

$$q_{\text{real}} = 2q \hat{\pi}_{\text{real}}, \quad q_{\text{fake}} = 2q (1 - \hat{\pi}_{\text{real}}), \quad (1)$$

which is *identical* to the symmetric rule at $\hat{\pi}_{\text{real}} = 0.5$ — so on a balanced pool the correction is provably inert and cannot be the source of any balanced-pool result — and shrinks the pseudo-real bucket to the estimated real fraction otherwise. The prior is estimated *once*, from the frozen pre-adaptation model: re-estimating each epoch lets a drifting model reinforce its own mistakes.

The obvious prior-free alternative is a fixed *confidence* threshold ($p \geq 0.95 \Rightarrow \text{fake}$) instead of a quantile. It does not remove the problem, it moves it: a quantile rule is calibration-free but assumes the prior, a confidence rule is prior-free but assumes calibration — and calibration is exactly the property the first paragraph of this section measures as *not* transferring across corpora. We run it as a baseline (Table I) rather than argue about it.

(c) Objective. With pseudo-labels $\tilde{y}$ from (b) on the confident set $\mathcal{C}$ and a stochastic channel perturbation $a(\cdot)$ (per-clip gain $g \sim \mathcal{U}(0.7, 1.3)$ plus Gaussian noise $\sigma=0.005$),

$$\min_{\{\gamma, \beta\}, \phi} \underbrace{\sum_{x \in \mathcal{C}} \text{CE}(f_\theta(x), \tilde{y})}_{\mathcal{L}_{\text{ST}}} + \lambda \underbrace{\|f_\theta(a(x)) - \text{sg}[f_\theta(x)]\|^2}_{\mathcal{L}_{\text{cons}}}, \quad (2)$$

optimising only the LayerNorm affine parameters $\{\gamma, \beta\}$ of the fine-tuned blocks and the head ϕ (16,706 parameters), with $\lambda=0.3$, $q=0.3$. Restricting the trainable set and using only prior-corrected confident tails is what prevents the degenerate collapse to which entropy minimisation is prone.

(d) A label-free guard. Adaptation needs a stop, and the step count has been the weakest-specified part of TTA recipes here. Since the source ranking is the asset being exploited, we monitor rank agreement with the frozen source model, $\rho_t = \text{Spearman}(s_{\theta_t}(\mathcal{T}), s_{\theta_0}(\mathcal{T}))$, and keep the last step with $\rho_t \geq \tau$ (Sec. IV).

III. EXPERIMENTAL SETUP

Protocol A (published baselines exist). The official ASVspoof2021-DF eval partition against the official CM keys: 400,435 of 611,829 trials are held locally and the missing part is uniform across codec, label and phase (verified 75–25 in every stratum), so the pooled EER is unbiased; true spoof rate 97.22%. Because our own source model is not competitive here (26.33% vs. a published 8.54%), we run the method on *released* checkpoints: DeepFense XLS-R 300M + AASIST trained on ASVspoof2019-LA [22], at three independent training seeds. Adaptation uses an unlabelled 20,000-clip subsample of the eval pool; we report the full official pooled EER and, as an inductive check, the EER over trials *not* in that subsample.

Protocol B (cross-corpus, balanced). Leave-one-corpus-out over four EER-capable targets: ASVspoof2019-LA [18]; D2 (LibriSpeech bona fide vs. eight open TTS systems); In-the-Wild [17]; and the Arabic Audio Deepfake set [20]. Each held-out target’s source pool merges the other three with a 38-language MLAAD subset [19]; held-out target audio is never in any source pool and six MLAAD languages are held out of every pool. Ten seeds per target per method.

Protocol C (a benchmark at its own prevalence). All 31,779 In-the-Wild clips at their native 37.2% fake rate — skewed the *opposite* way to DF’s 97.2%.

Comparison points. Source-only; the median-threshold rule; Tent, SHOT, ETA/EATA, SAR; AdaBN [14]; DANN [13] (trained to convergence with the held-out target’s unlabeled inputs — more privileged access than ours); an ASDG-style baseline [5]; and a target-supervised oracle. Two published constants in the modern TTA baselines are degenerate at $C=2$ and were re-derived rather than transplanted. EATA’s redundancy filter ($|\cos(p, \bar{p})| < 0.05$) is calibrated for $C=1000$; on the binary simplex the minimum attainable cosine is 0.71 at $\bar{p}=(.5, .5)$ and 0.11 at $(.9, .1)$, so nothing passes until the running average itself reaches a vertex, after which only *minority*-predicted samples pass — neither is the stated mechanism, so we run the reliability filter alone, which is EATA’s own published ablation. SAR’s recovery trigger $e_0=0.2$ sits far below ImageNet’s $E_0=0.4 \ln 1000=2.76$ but *inside* the $C=2$ reliable band $H < 0.277$, firing on every batch and freezing the model at its initial weights (verified: 32 resets in 32 steps, zero net movement); we keep its relative position, $e_0 = (0.2 / \ln 1000) \ln C = 0.020$.

TABLE I

OFFICIAL ASVSPOOF2021-DF EVAL, 400,435 TRIALS, 97.22% SPOOF, ON A released XLS-R+AASIST CHECKPOINT [22] WE DID NOT TRAIN. ALL ADAPTATION IS LABEL-FREE. “DEFICIT” IS $(100 - \text{EER}) - \text{acc}@0.5$: THE ACCURACY THE RANKING SUPPORTS BUT THE THRESHOLD DOES NOT DELIVER. ρ (RANK AGREEMENT WITH THE FROZEN SOURCE MODEL) AND THE PREDICTION-RATE GAP ARE THE TWO LABEL-FREE MONITORS OF SEC. II(D) — COMPUTABLE AT DEPLOYMENT, UNLIKE EVERY OTHER COLUMN.

	EER %	AUC	acc@0.5	deficit	ρ	gap
Source (as released)	4.51	.9923	98.87	-3.38	1.000	.000
+ median-threshold rule	4.51 [†]	.9923 [†]	59.06	—	—	—
+ ETA [8]	7.44 [‡]	.9288	98.93	-6.37	0.312	.007
+ Tent [7]	4.06 [‡]	.9765	98.52	-2.58	0.438	.013
+ SHOT [10]	6.38	.9725	77.64	+15.99	0.767	.225
+ TTA, fixed confidence 0.95	[8b]					
+ TTA, symmetric $q=0.3$	5.84	.9846	62.96	+31.20	0.816	.372
+ TTA, prior-corrected (disjoint/inductive half)	4.02	.9931	98.95	-2.97	0.867	.007
+ SAR [9]	4.47 [§]	.9923	98.88	-3.35	0.970	.000

published Protocol-A systems, for scale: Wav2DF-TSL 1.95; SSL+SLS 2.87; Wav2Vec2-AASIST 8.54; challenge DF top-1 15.64; our own source model 26.33.

[†] A threshold cannot change EER or AUC; both come from the score ROC. The median rule is a control for the *accuracy* claim and cannot produce any EER result in this paper. [‡] EER over a pool that is more than 93% a single tied score (Tent saturates 98.07% of scores to exactly 1.0, ETA 97.71%); read AUC. [§] SAR neither damages nor adapts: 308 recovery resets in 2,500 steps, AUC unchanged to four decimals.

Baseline fairness. Every TTA arm adapts the same 16,706 parameters with the same lr, batch and pass count. None was tuned separately, and none could be: per-target TTA tuning needs target labels, the one resource the setting assumes absent.

Implementation. 16 kHz audio, top-4-layer XLS-R fine-tuning for our own source models (Adam, $\text{lr} \times 10^{-4}$, batch 32, 8 epochs). Adaptation: $\text{lr} \times 10^{-4}$, $E=4$ passes, batch 32 (third-party crops matched to each checkpoint’s own training length). Code and manifests: [URL withheld].

IV. RESULTS

A. The class prior is the hidden variable

Table I is the main result. Applied unmodified to a released checkpoint that is already better than the published Wav2Vec2-AASIST baseline on this protocol, the symmetric-quantile configuration *damages* it: accuracy $98.87 \rightarrow 62.96\%$, EER $4.51 \rightarrow 5.84\%$, and it manufactures a 31-point calibration deficit where there was none. The mechanism is mechanical: $q=0.3$ labels the bottom 30% of a 97.2%-spoof pool “confidently bona fide”.

The published baselines are no better placed, and one states the assumption outright: SHOT minimises $\mathbb{E}[H(p)] - H(\mathbb{E}[p])$, whose diversity term is maximised when the mean prediction is *uniform* — at $C=2$ a hard-coded prior of $\frac{1}{2}$, asking it to move 47 points of probability mass into the bona fide class. It degrades the checkpoint accordingly, with no score saturation to blame. The regulariser introduced to *cure* Tent-style collapse encodes the assumption more literally than Tent does.

ETA fails more instructively. Its reliability filter exists to discard the high-entropy samples that destabilise entropy minimisation; here it admitted 99.8% of them, so ETA reduced to Tent and landed worse (AUC .9288, the largest ranking loss of any arm). That is structural: an entropy-keyed filter separates confident from uncertain, not *confident-and-right*

from *confident-and-wrong*, and mis-calibration manufactures exactly the latter. A safeguard keyed on the model’s own confidence inherits the broken calibration it guards against.

SAR is the exception that clarifies the rule. It is the only prior method here that does not degrade the checkpoint — and it manages that by declining to move it: 308 model-recovery resets in 2,500 steps, an AUC unchanged to four decimals, and a score distribution indistinguishable from the source model’s. Its collapse guard does exactly what it promises, and prevention is all it delivers. So the claim is not that every published method fails here. It is that *three of the five prior configurations degrade a state-of-the-art detector at the standard benchmark’s own prevalence, the fourth avoids damage only by reverting its own updates, and none of them improves it.*

Replacing the symmetric budget by Eq. (1) with a label-free prior estimate reverses this. BBSE returns $\hat{\pi}_{\text{fake}}=0.9715$ against a true 0.9708, giving a budget of (0.017, 0.583) instead of (0.3, 0.3), and EER improves to 4.02% — with no target labels and no retraining. The change between the two arms is one scalar.

Replication over three independently trained checkpoints. EER % and accuracy at $\tau=0.5$, same protocol, same pool:

	seed 2	seed 42	seed 240
source	4.51 / 98.87	4.56 / 98.78	3.82 / 99.11
+ symmetric q	5.84 / 62.96	4.08 / 61.80	4.66 / 65.24
+ prior-corrected	4.02 / 98.95	3.29 / 98.98	3.80 / 98.74

Two things replicate and one does not, and the distinction is the thesis restated. The *operating-point* collapse under a symmetric budget replicates on all three (accuracy 62–65%, deficit +30 to +34), and prior correction beats the symmetric budget on all three (−1.82, −0.79, −0.86 EER). What does *not* replicate is that budget’s effect on *ranking* — EER worsens on two checkpoints and improves on one. A mis-specified prior reliably breaks the operating point, not the ranking. Against source-only, prior correction improves EER on 3/3 but by very different amounts (1.27, 0.49, 0.02; AUC up on two, flat on the third), so we rest on the unanimous controlled comparison and report the against-source magnitudes as varying.

B. Why this is not a re-thresholding result

Both EER and AUC improve, and neither takes a threshold as input, so no operating-point rule can produce this gain. The cleanest form of that argument uses the *same* prior estimate twice: re-point the threshold with it and take no gradient step, and accuracy moves $98.92 \rightarrow 98.94\%$ while EER stays at 4.51 by construction; feed it to the pseudo-label budget instead and EER goes to 4.02. The prior does its work in the budget, not at the threshold.

On *balanced* pools a one-line median-threshold rule is genuinely competitive with gradient adaptation on accuracy — we reported it as such, and it is the right control for a calibration claim. Resampling the DF scores across target priors shows why, and what it costs: the median rule gives

TABLE II
CROSS-CORPUS EER % / AUC, MEAN OVER 10 SEEDS
(LEAVE-ONE-CORPUS-OUT, BALANCED POOLS, TRANSDUCTIVE).
DROPPING THE CONSISTENCY TERM ($\lambda=0$) GIVES
4.20/33.39/11.55/22.19.

Method	ASVspoof2019	LibriSpeech-TTS	In-the-Wild	Arabic
Source-only	5.03 / .989	33.54 / .727	12.78 / .945	22.50 / .843
Tent [7]	23.47±21.96 / .776	35.44 / .674	27.34±11.21 / .735	47.35±5.03 / .527
SHOT [10]	[s4]	[s4]	[s4]	[s4]
ETA [8]	[s4]	[s4]	[s4]	[s4]
SAR [9]	[s4]	[s4]	[s4]	[s4]
DANN [13]	7.35 / .966	34.43 / .717	13.34 / .924	28.53 / .759
ASDG [5]	4.98 / .987	32.64 / .749	17.33 / .891	27.41 / .798
Ours	3.47 / .994	33.74 / .727	11.33 / .958	20.99 / .866
Oracle (target-supervised)	0.89 / .999	13.93 / .939	4.42 / .992	9.50 / .968

95.6% accuracy at a prior of 0.5 (shipped threshold 88.3, labelled oracle 95.7) and falls monotonically to 59.0% at the benchmark’s own 0.972, while a threshold at the $(1-\hat{\pi})$ quantile tracks the shipped one throughout.

That is the rule’s assumption, not an accident — and every cell on which it was previously evaluated satisfied that assumption by construction, having been balanced to a prevalence of 0.489–0.539 by the evaluation pipeline itself.

C. Cross-corpus, balanced pools

Here the prior correction is inert by construction ($\hat{\pi} \approx 0.5$ returns the symmetric budget exactly), so Table II measures the rest of the method. It wins on three of four targets (10/10 seeds on ASVspoof2019 and Arabic, 8/10 on In-the-Wild) and is EER-neutral on LibriSpeech-TTS, the target with the lowest source AUC (0.73). Per-target Wilcoxon against source-only is significant on ASVspoof2019 and Arabic ($p=0.0020$, Holm 0.0078) and nominal on In-the-Wild (0.0293, Holm 0.0586). Against the modern TTA baselines, [stage 4].

Tent is unpredictably catastrophic, and sometimes silently so. On ASVspoof2019 its per-seed EER splits five seeds near source-only and four at or near chance with no warning sign (std 21.96); on Arabic it is a consistent chance-level failure (9/10 seeds). Ours shows no comparable collapse anywhere (largest std 2.01). The failure is a ranking failure first: on the strongest and weakest public checkpoints Tent collapses to chance on one (16.41 $\rightarrow$ 49.80% EER, AUC .918 $\rightarrow$.502) but only degrades AUC on the other. On the DF pool the damage does not show in the field’s headline metric at all — Tent returns a *lower* EER than source (4.06 vs. 4.51) while saturating 98% of the pool to one tied score. A TTA study reporting EER without a score-resolution check can record a collapse as an improvement.

D. Label-free guards

If the source *ranking* is the asset adaptation exploits, an update that destroys it has discarded its own precondition — and that is visible without labels, by comparing the adapted scores’ *order* against the frozen source model’s (ρ , Table I). A second monitor, the drift of the predicted positive rate from $\hat{\pi}$, sees the other failure. Over the eight adapted arms we ran on this pool the two are complementary and both are needed: ρ catches ranking collapse decisively (ETA 0.31, Tent 0.44) but is blind or marginal on the operating-point collapses, while the gap catches those decisively (0.37, 0.39, 0.23) and is blind to

ranking collapse (0.007 for ETA). Requiring both — $\rho \geq \tau$ and $\text{gap} \leq 0.05$ — separates the five harmful arms from the three harmless ones exactly, for any τ in $[0.70, 0.80]$.

That plateau matters more than the exactness, because the exactness is fragile: an earlier ρ -only rule at $\tau=0.85$, fitted on the first seven arms, looked equally exact and was then falsified by the eighth — it rejected the best result in this study (seed 42, $\rho=0.828$, EER 4.56 $\rightarrow$ 3.29). Two thresholds fitted on eight points is a diagnostic, not a validated rule. [stage 6/7] gives the out-of-sample version: per-epoch traces on the four leave-one-corpus-out targets, thresholds chosen leave-one-target-out against the oracle budget, with churn and confidence as controls.

E. Breadth, and the mechanism

Applied unmodified to nine released checkpoints across five architectures and ten corpora (369 cells), the consistent effect is *re-calibration*: δ falls 7.29 $\rightarrow$ 1.96 points (248/369, $p=2.2 \times 10^{-17}$), with $\text{corr}(\delta_{\text{before}}, \delta_{\text{after}}) = +0.17$ — where it lands barely depends on where it started — and the same rule designed against XLS-R re-calibrates a mel-spectrogram transformer [16] unchanged (39/39, $p=5 \times 10^{-8}$). The mechanism is *not* learned invariance: a proxy $\mathcal{A}$ -distance and a linear domain-origin probe both *rise* after adaptation (Arabic 1.545 $\rightarrow$ 1.606), the opposite of what domain-invariance predicts and consistent with DANN underperforming while optimising that very quantity. What the gradient adds beyond any threshold is separability where the ranking is strong: on ASVspoof2019 it raises within-attack-family AUC on all six families (30/30, $p=10^{-9}$); on AUC-0.71 LibriSpeech-TTS it is null.

V. LIMITATIONS AND CONCLUSION

- **Prior estimation assumes label shift; corpora also shift covariates.** Every estimator we tried shares one positive bias — $\hat{\pi} \approx 0.61$ at a true 0.50, 0.974 at a true 0.972 — because DF’s codecs raise the detector’s real-class error above its source-domain rate. A bias shared by four unrelated estimators belongs to the shift, not the estimator, and it is smallest where it matters most.
- **The prior fixes the budget, not the threshold.** Where a shipped threshold is near-optimal the BBSE threshold buys nothing over $\tau=0.5$ (98.94 vs. 98.92); the gain comes from the budget, and its size varies a great deal across checkpoints (1.27 to 0.02 EER points).
- **Needs a reliable source ranking, and improves someone else’s model.** A low-AUC start is not a regime we can rescue (the AUC-0.71 target is the one null in Table II), and we did not train a Protocol-A-competitive detector ourselves (26.33% vs. a published 8.54%), so every claim is of the form “this wrapper improves that detector”.

Cross-corpus deepfake detection fails at the operating point, and the repair needs no labels — but the recipes used to make it assume a class-balanced pool, which holds where TTA is measured and essentially nowhere else. A label-free prior estimate removes it, and is provably inert at $\hat{\pi}=0.5$.

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